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Portable AI-powered scanning slit-light device for low-cost eye disease screening

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Abstract

Accurate anterior-segment assessment is essential for detecting conditions such as primary angle-closure glaucoma, cataract, and keratoconus, yet current tools remain limited: slit-lamp biomicroscopy is qualitative and operator-dependent, whereas anterior segment OCT (AS-OCT) is costly and clinic-bound. We introduce an AI-integrated portable scanning slit-light device that delivers quantitative anterior-segment biometry at a material cost below USD 500. The system combines a motorized slit-scanning mechanism with synchronized imaging and an on-device deep-learning model (LWBNA-unet) to segment corneal and iris reflections, pupil boundaries, and corneal surfaces. Geometry-aware corrections—including slit-incidence compensation and per-frame anatomical scaling—enable calibrated estimation of anterior chamber depth (ACD) as the primary quantitative output, with exploratory estimates of central corneal thickness (CCT) derived from the same scan. In a clinical study of 170 participants, ACD showed excellent agreement with AS-OCT (Pearson's $r \approx 0.92$, concordance correlation coefficient ≈ 0.90 ; mean bias ≈ 0.0 – 0.04 mm with 95% limits of agreement LoR $\sim \pm 0.3$ mm) indicating near-clinical interchangeability for ACD in a screening context. Representative cases illustrated clear visualization of anterior-segment features associated with narrow angles, cataract, corneal opacity, and keratoconic ectasia from a single scanning-slit video. A typical 51-frame video (captured in ~ 15 sec) can be fully processed on a Jetson Orin

Nano in 18.5 s (≈ 2.7 fps), supporting compact, battery-powered deployment. These results establish the first ultra-low-cost platform capable of automated, quantitative, and anatomically calibrated anterior-segment imaging, offering a scalable foundation for community screening and teleophthalmology in resource-limited settings.

Introduction

Vision impairment affects more than two billion people worldwide, and its prevalence is projected to double within the next three decades. A large proportion of avoidable blindness arises from diseases that primarily involve anterior segment of the eye, including cataract, corneal opacity, and primary angle-closure disease.¹ The burden is disproportionately higher in low- and middle-income countries, where access to eye care is up to four times lower than in high-income regions.² Even in developed nations, cost, access, and awareness remain major barriers: in the United States, for instance, millions of adults at risk of visual disability do not receive regular examinations.^{3,4} Early detection and intervention can prevent up to half of all vision loss, yet current ophthalmic infrastructure remains resource-intensive and fragmented. A comprehensive evaluation often requires multiple instruments—slit-lamps, tonometers, pupillometers, fundus cameras, and OCT scanners—each operated by trained personnel.⁵⁻⁸

Slit-lamp biomicroscopy remains the cornerstone of anterior-segment evaluation, providing detailed visualization of the cornea, anterior chamber, iris, and lens. However, conventional slit-lamps are bulky, immobile, and cost several thousand U.S. dollars, confining their use to clinical settings. While portable and handheld variants have been introduced,⁹⁻¹⁷ no commercially available slit-lamp—tabletop, portable, or smartphone-based—provides automated, quantitative anterior-segment measurements. All existing systems remain fundamentally qualitative, relying on operator interpretation or low-precision indirect methods, with no integrated AI for objective measurement or analysis. Anterior-segment optical coherence tomography (AS-OCT), while quantitative and reproducible, is even more cost-prohibitive, with typical system prices exceeding USD 50,000, making it impractical for community screening or resource-limited settings.¹⁸⁻²⁰ This cost-access gap highlights the urgent need for a low-cost, automated, and deployable alternative capable of providing reliable anterior-segment measurements and AI-based assessment of anterior-segment anatomy and disease risk.

Here we present a standalone, AI-integrated portable scanning slit-light device that overcomes the limitations of conventional and smartphone-based systems. The compact system (< USD 500) combines a motorized slit-scanning mirror with onboard AI (LWBNA-unet^{21,22}) for real-time segmentation of corneal and iris reflections, pupil boundaries, and corneal surfaces. It provides calibrated quantitative measurements together with disease-related features such as cataract opacity, corneal ectasia, and narrow anterior-chamber angles associated with increased

risk of primary angle-closure glaucoma. To our knowledge, this is the first ultra-low-cost, edge-AI slit-lamp quantitatively validated against AS-OCT, establishing a scalable foundation for community-level anterior-segment screening and teleophthalmology.

Materials and Methods

This prospective clinical study evaluated the reliability and validity of images acquired using a standalone, AI-enhanced portable scanning slit-lamp device for automated anterior-segment analysis. The study protocol was approved by the Clinical Research Ethics Committee of Tohoku University, Graduate School of Medicine (No 2025-1-352) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants and/or their legal guardians prior to enrollment. The study was registered in the UMIN Clinical Trials Registry (UMIN000036747). Although the imaging device was developed within this study, its use was limited to non-invasive image acquisition and did not alter patient management, diagnosis, or treatment plans. Accordingly, the study is classified as a non-interventional observational clinical research study, with no assignment of interventions, randomization, or modification of standard clinical care.

Participants were recruited during routine outpatient visits to Tohoku University Hospital and from the general public during the annual *Glaucoma Awareness Week* in Sendai, Japan. Eligibility criteria included adults aged 18 years or older who were able to provide informed consent.

Enrollment was independent of ocular disease status to ensure representation of both healthy and pathological eyes. In total, approximately 170 participants were enrolled, and both eyes were imaged using the prototype device. Images in which anterior-segment structures were poorly visualized were automatically excluded by the device's quality-control algorithm. Of the acquired dataset, approximately 50 eyes with corresponding AS-OCT scans were used for quantitative validation, while the remaining images were utilized for deep-learning model training and optimization. Mean age of the subjects was 65.68 ± 8.76 years, with 21 of 50 participants (42%) males and 29 (58%) females. Spherical equivalent (SE, by eye) showed a predominantly myopic profile, with mean $SE \approx -2.52 \pm 3.17$ D (median -2.44 D; range -9.62 to $+7.38$ D). Based on SE, 15.2% of eyes were classified as high myopia (≤ -6.00 D), 60.9% as myopia (-6.00 to -0.50 D), 9.8% as emmetropia (-0.50 to $+0.50$ D), 12.0% as hyperopia ($\geq +0.50$ to $< +5.00$ D), and 2.2% as high hyperopia ($\geq +5.00$ D). Visual acuity was generally good, with a mean logMAR of 0.04 ± 0.30 (median -0.08 ; range -0.18 to 2.00), indicating that most eyes were corrected to approximately 20/20 or better. AS-OCT imaging was performed using the CASIA-2 Advance system (Tomey Corporation, Nagoya, Japan), with pupil-centered vertical meridian scans (90° – 270°) acquired by certified optometrists under standardized protocols. For each participant, at least two high-quality scans were obtained.

Design of scanning-slit light device: Figure 1 illustrates the optical geometry of the proposed system. The portable scanning slit-lamp system consists of an imaging unit integrated with a control module. The

illumination source is a white LED, collimated through convex lenses and a 400 μm slit aperture. In the present prototype, this slit width is set by a fixed mechanical aperture and is not adjusted during acquisition, and it was chosen empirically as a compromise between a sufficiently thin optical section for axial discrimination, adequate signal-to-noise ratio for segmentation, and minimizing patient discomfort from excessive light intensity. A mirror redirected the optical path to minimize device size, while a servo-motor-driven scanning mirror projected the slit beam across the eye. The camera optical axis is fixed and observes the anterior segment under a moderate oblique angle of approximately 24° relative to the corneal surface normal. We define θ as the illumination incidence angle at the eye, measured relative to the local corneal surface normal (and not as the mechanical rotation angle of the scanning mirror). At the beginning of the scan, the slit is positioned at one peripheral corneal edge with $\theta_0 = 63^\circ$, corresponding to a classical Van Herick configuration ($\sim 60^\circ$). As the mirror rotates, the slit sweeps laterally across the cornea and limbus, and at the opposite edge reaches $\theta_1 = 47^\circ$. During this process, mirror rotation translates the illumination position across the ocular surface but does not rotate the camera. Because the camera axis remains fixed, the

illumination–observation separation angle varies smoothly across the scan (as a geometric consequence of the changing illumination incidence). Although the illumination angle changes during scanning, the effective observation geometry is quasi-fixed. This is confirmed experimentally by fitting an ellipse to the corneal boundary in each frame. Across frames, the ellipse axis ratio remains high (typically 0.9–0.95 when the full cornea is visible), indicating that the camera observes the cornea under a relatively stable oblique angle of approximately $20\text{--}30^\circ$. Small variations are attributable to natural eye position variability and perspective effects, but no large changes in viewing direction occur during scanning. Therefore, angular correction is critical for corneal thickness measurements derived from slit broadening, whereas anterior chamber depth, defined by the separation between two high-contrast reflective interfaces is substantially, less sensitive to the scanning geometry.

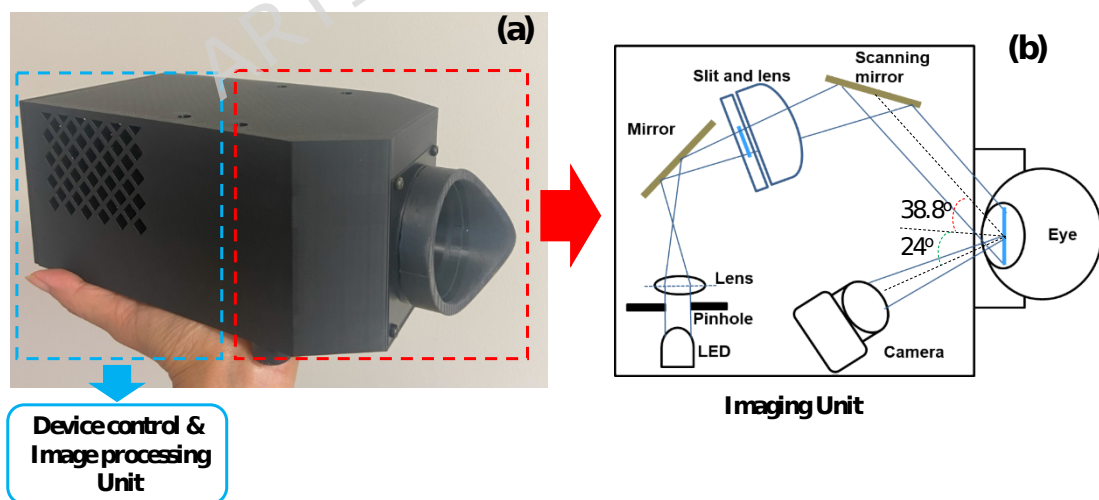


Figure 1. Portable scanning slit-light device and optical imaging architecture. (a) Photograph of the handheld device, (b) Schematic of the imaging unit illustrating the illumination path, slit and lens configuration, motorized scanning mirror, and camera geometry used to generate calibrated

The scanning mirror was controlled by an NVIDIA Jetson Nano developer kit, ensuring precise synchronization with image capture. Images were recorded using a programmable compact camera (IMX477 CMOS sensor, 4056×3040 pixels, 20–30 fps) coupled to an M12-mount lens (focal length ≈ 3.9 mm, horizontal field of view $\approx 75^\circ$) with electronically adjustable focus. Raw frames were acquired at full sensor resolution and subsequently cropped to a region of interest centered on the anterior segment to remove background regions and improve processing efficiency. Cropping does not alter the physical pixel size but limits the field of view to the cornea, iris, and lens. For real-time visualization and computational stability, frames were further down-sampled by a fixed scaling factor prior to analysis; this controlled resizing changes the effective pixel sampling density but preserves all geometric relationships within the image. Based on the measured corneal diameter in pixel space (~ 300 pixels corresponding to ~ 12 mm), the effective lateral sampling resolution in the corneal plane is approximately 35–45 μm per pixel. This object-space resolution reflects the combined effects of lens focal length, working distance (~ 30 mm), oblique viewing geometry, and image resizing, rather than the native sensor pixel pitch (1.55 μm). The slit width was kept fixed throughout acquisition to ensure stable illumination geometry and consistent geometric correction across frames. Camera acquisition and mirror scanning were tightly synchronized by the Jetson Nano board to ensure accurate spatial sampling. All examinations were conducted in a darkened/low-light environment to minimize external light interference and enhance contrast of corneal and lenticular

reflections. For most participants, vertical slit-scan videos were captured using the portable device [Fig. 2(a)]. The system can also operate in a horizontal scanning mode simply by rotating the device 90° , as illustrated in [Fig. 2(d)]. This rotation changes only the scan orientation (vertical vs. horizontal) relative to the eye and does not alter the underlying optical geometry or angular relationships between the illumination and camera axes. The device is handheld and gently stabilized against the orbital rim without a chin or forehead rest, consistent with its intended use in portable screening environments. Participants were instructed to fixate on the device's built-in target while the operator monitored the live video feed to maintain alignment during scanning. To mitigate motion artifacts and

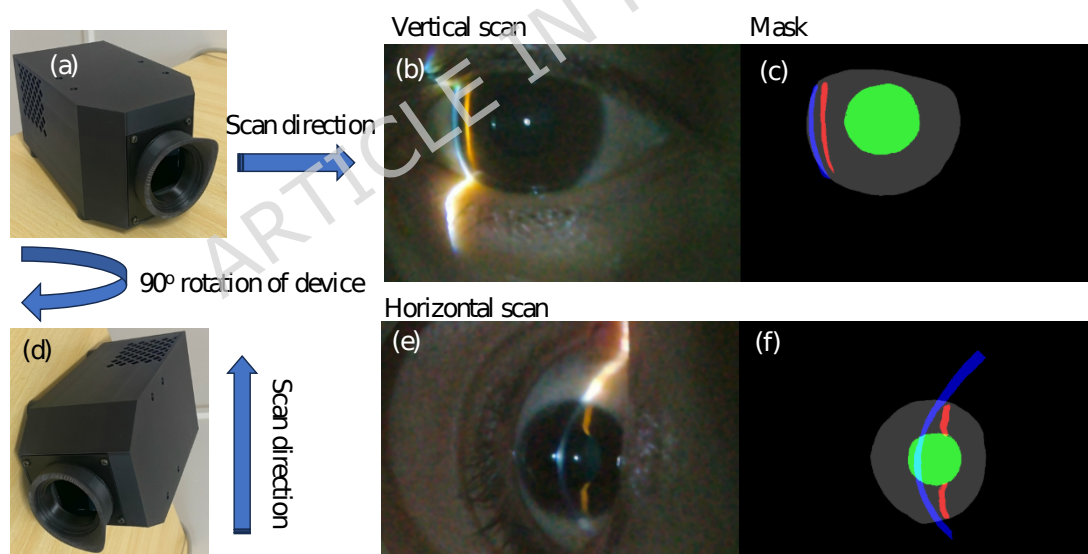


Figure 2. Imaging configurations and representative outputs of the portable scanning slit-light device. (a) Portable scanning slit-lamp device positioned for vertical-mode imaging. (b) Representative frame from a normal eye showing slit reflections on the corneal (blue) and iris (orange) surfaces. (c) Corresponding segmentation mask delineating corneal reflections (blue), iris reflections (red), pupil boundary (green), and outer corneal surface (gray). (d-f) Horizontal-mode imaging setup and representative results analogous to (a-c),

blinks, slit-scan videos were acquired over approximately 15 s using bidirectional scanning, yielding 50–70 frames per eye. After acquisition, all frames were processed with the LWBNA-unet segmentation model, which outputs masks for corneal and iris reflections, the pupil boundary, and the anterior and posterior corneal surfaces. Frames were retained for biometric analysis only if all required anatomical landmarks were successfully segmented; frames lacking visible corneal reflection (e.g., when the slit beam was on the sclera) or affected by blink- or motion-induced blur, in which the model failed to delineate reflections and pupil boundaries, were automatically discarded. This segmentation-based acceptance criterion served as the primary quality-control mechanism, rather than explicit thresholds on image sharpness, brightness, or entropy parameters.

A subset of participants was imaged in both orientations to expand the diversity of training frames for the deep-learning segmentation model. For reference, conventional imaging was performed using a standard slit-lamp (Topcon SL-D701) and anterior segment optical coherence tomography (AS-OCT) (CASIA-2, Tomey Corporation). After each imaging session, participants provided structured feedback on device comfort, usability, and overall experience.

Imaging under low-light conditions: Reflected slit-light intensity is inherently weak due to high transparency of corneal and lenticular tissues. High illumination improves structural visibility but reduces patient comfort, while lower intensity increases noise. To balance image quality

and comfort, we implemented a synchronized video acquisition pipeline (Figure 3). At each mirror angle A (slit position x_i), 4–10 frames were captured. The first 2–3 frames were discarded to allow system stabilization,

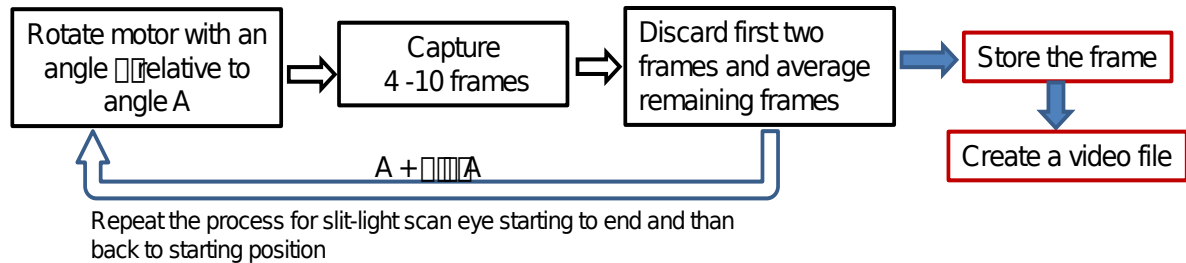


Figure 3. Workflow for video acquisition using the portable scanning slit-light device.

Illustrating the integrated capture pipeline of the AI-enhanced portable slit-light system. The process begins with motorized slit scanning and synchronized frame and the remaining frames were averaged into a single high-quality image. The mirror was then rotated by a small step ΔA , and the process repeated until the beam traversed the full ocular span (in less than 15 secs). A complete scan generated ~50–70 averaged frames, yielding compact but diagnostically rich datasets. This method provided clear visualization of corneal reflections, iris, lens surfaces, and pupillary boundaries under low-intensity illumination, while reducing file size compared with conventional video recordings.

Automatic processing of device video with lightweight deep learning model (LWBNA-unet):

Automated feature extraction provides clinically relevant parameters that are traditionally assessed manually by ophthalmologists, thereby enabling objective, scalable, and real-time diagnosis within a portable standalone

system. All captured frames were processed using our lightweight deep learning model, LWBNA-unet.²¹ The architecture is derived from a U-Net backbone, augmented with channel-attention modules in the encoder, decoder, and bottleneck pathways. Successive narrowing of feature channels in the bottleneck reduced computational load while maintaining segmentation performance. This design enabled high segmentation accuracy with minimal overhead, making the model suitable for deployment on resource-constrained platforms such as smartphones, Raspberry Pi, and NVIDIA Jetson devices.

The anatomical features targeted for segmentation included corneal reflections, iris reflections, pupil boundaries, and the outer corneal surface boundary. For training and validation, ~1,600 image frames were extracted from ~100 videos captured with a portable scanning slit-light device. These training and validation videos did not include any eyes used for quantitative AS-OCT validation in the present study. Ground-truth masks delineating these features were manually annotated. Representative examples of segmentation masks are shown in Figures 2(c) and 2(f), illustrating results in both vertical and horizontal scanning modes. The dataset was divided into 80% for training and 20% for validation. Model training was performed in a supervised manner using paired images and ground-truth masks resized to 640×320 pixels. Optimization employed the Adam optimizer (initial learning rate 8×10^{-4} , $\beta_1 = 0.5$) with an exponential learning-rate decay (decay rate 0.9). A weighted Dice loss function was used to mitigate class imbalance among the segmented anatomical structures. Training was conducted with a batch size of 4 for

up to 300 epochs, with early stopping applied if validation loss failed to improve for 100 consecutive epochs. To improve robustness and generalization, on-the-fly data augmentation was applied to input images only, including contrast-limited adaptive histogram equalization (CLAHE), random grayscale conversion, intensity normalization or clipping, and additive noise with randomly scaled amplitudes. The final model produced four output channels, each corresponding to one anatomical feature of interest.

Before estimating quantitative parameters such as anterior chamber depth (ACD) and central corneal thickness (CCT), frames of interest were automatically extracted from the portable scanning slit-light device videos (an example of captured and processed video is shown in Supplementary Video 1). Accurate parameter estimation required frames in which the anatomical features were optimally visualized. For example, CCT estimation relied on frames where the corneal reflection was aligned with the optical axis, while ACD estimation required frames in which the iris reflection was centered relative to the pupil. To achieve this, each video was segmented into anatomical components using the trained LWBNA-unet model, and candidate frames were ranked according to quantitative criteria derived from the segmented contours. For each segmented frame, the pupil center was approximated by the image center after alignment, and the corneal and iris reflection centers were computed from their respective segmented contours. The frame in which the corneal reflection was closest to the pupil center in terms of minimal horizontal displacement was selected as the cornea-centered frame, while the frame with the iris

reflection minimizing horizontal displacement relative to the pupil center was designated as the iris-centered frame (Supplementary Video 1). ACD and CCT were computed from the iris-centered and cornea-centered frames, respectively, as described below. In the cornea-centered frame, the corneal apex was identified along the reconstructed corneal reflection centerline as the point at which the local slope changed sign, corresponding to the transition across the optical axis. Central corneal thickness (CCT) was measured at this apex as the perpendicular distance between the anterior and posterior corneal boundaries using locally estimated normal vectors. The same corneal apex detection was re-applied in the iris-centered frame to ensure consistency, and ACD was computed as the distance between the corneal apex and the iris reflection centroid, with half of the local corneal thickness subtracted to account for corneal overlap. A separate scoring metric incorporating mean pixel intensity, standard deviation, and coverage ratio within the pupil mask was used to identify the lens opacity frame, ensuring robust analysis of lens features. Lens density was quantified from this frame as the mean grayscale intensity within the masked pupil region. Finally, the iridocorneal angle (ICA) is estimated from the iris-centered frame by extracting the lower corneal boundary and upper iris boundary from the segmentation output, reconstructing each boundary as a continuous curve, and fitting local linear segments near their point of closest approach. The acute angle between these fitted segments is then computed to yield quantitative ICA measurements. Anterior chamber area is estimated by integrating the

enclosed region between the posterior corneal boundary and the anterior iris boundary within the segmented frame.

Frames in which corneal, iris, or pupil reflections were not detected—due to motion artifacts, defocus, or other causes—were automatically excluded by the algorithm. For the 50 eyes included in this study for quantitative validation of ACD and CCT, an average of 24.3 ± 4.5 frames per scan (median 25; range 12–32) out of 51 were retained as “good frames.” This corresponds to an approximate retention rate of 48% (range ~24–63%) per video. The excluded frames were typically at the beginning or end of the sweep (slit on sclera) or during blinks and brief motion (Supplementary Video 1). All scans yielded a sufficient number of accepted frames for downstream ACD and CCT estimation.

Geometric correction and measurements of corneal Thickness and distance from image frame: Because the slit beam falls obliquely on the corneal surface (under a Van Herick-like illumination geometry Fig. 1), the reflected light band appears broadened in the captured frames, resulting in a systematic overestimation of thickness. To address this, a geometric correction was applied based on the illumination incidence angle at the eye. The angle at any horizontal position x across the image was interpolated as:

$$\theta(x) = \theta_0 + \frac{(\theta_1 - \theta_0)x}{w} \quad (1)$$

where $\theta_0 = 63^\circ$ corresponds to the initial slit position at the peripheral corneal edge ($\approx 60^\circ$ Van Herick configuration) and $\theta_1 = 47^\circ$ corresponds to the slit position at the opposite side after scanning, x is the horizontal

pixel position, and W is the image width (460 pixels). The corrected corneal thickness $[T_{true}(x)]$ was then estimated as:

$$T_{true}(x) = T_{measured}(x)\cos(\theta(x)) \quad (2)$$

where $T_{measured}$ represents the apparent slit thickness observed in the image. This correction compensates for first-order geometric slit broadening caused by oblique illumination. Central corneal thickness (CCT) was evaluated in the cornea-centered frame, with $T_{measured}$ obtained at the corneal apex.

To convert thickness values from pixels to microns, the outer corneal contour in the same frame was fitted with an ellipse, and the major axis length was used to estimate the corneal diameter in pixels. Because the camera observation geometry remains fixed during slit scanning, the corneal contour is subject to the same magnification effects as the thickness measurement. Assuming a standard anatomical diameter of 12 mm, a per-frame scaling factor was derived and applied. The final CCT values were reported in microns after both geometric correction and anatomical scaling. It is to be noted that this geometric model does not explicitly correct for refraction at ocular interfaces; all distances are measured along the camera observation direction in image space and should be interpreted as effective geometric surrogates rather than ray-traced optical path lengths. Under the moderate oblique angles used and the relatively short intraocular distances involved, the resulting systematic error appears small, as quantified in the method-comparison analysis against AS-OCT.

Anterior chamber depth was defined as the distance between the corneal endothelium and the anterior iris surface measured along the camera observation direction. In segmented frames, the corneal reflection center and iris reflection center were automatically identified. The pixel distance between these two points was first measured and then corrected by subtracting half the CCT (to account for the posterior corneal boundary). Because both interfaces are observed under the same fixed viewing geometry, relative depth measurements are largely insensitive to slit incidence angle. Conversion into physical units was performed using the same 12-mm scaling factor.

$$ACD = \left(d_{ci} - \frac{1}{2} CCT \right) \cdot \frac{12 \text{ mm}}{D_{cornea}} \quad (3)$$

where d_{ci} is the distance between the corneal and iris reflection centers in pixels, CCT is the corrected corneal thickness, and D_{cornea} is the corneal diameter in pixels obtained from ellipse fitting.

Dynamic scaling based on the white-to-white corneal diameter was deliberately chosen over grid-based or fixed-distance calibration methods to ensure robustness against subject motion in portable screening environments. During acquisition, lateral and vertical eye motion (x-y shifts) primarily introduce translational displacements in the image plane without altering magnification. These shifts are effectively compensated by segmentation-based alignment, in which frames are re-centered using the pupil center, thereby removing x-y positional variability prior to biometric analysis. In contrast, axial eye motion along the optical axis (z shifts), caused by forward or backward head movement in the absence of

chin or forehead rests, leads to changes in effective magnification while preserving relative image geometry. Such z-dependent magnification changes cannot be corrected by fixed grid-based calibration. Instead, scaling measurements using the segmented white-to-white corneal diameter provides a self-consistent anatomical reference that undergoes the same magnification effects as corneal thickness, anterior chamber depth, and related parameters.

Statistical analysis. Agreement between biometric measurements obtained with the proposed portable scanning-slit device and the reference instrument (CASIA-2, AS-OCT) was evaluated using standard method-comparison statistics. Continuous variables are summarized as mean $\pm$ standard deviation (SD). Bland-Altman analysis²³ was used to calculate the mean inter-device difference (bias), SD of the differences, and 95% limits of agreement (LoA = bias $\pm$ 1.96 $\times$ SD). The LoA width was also expressed as the coefficient of repeatability (CR = 1.96 $\times$ SD), representing the range within which most inter-device differences are expected for an individual eye.

Pearson's correlation coefficient (r) quantified the linear association between devices. To assess calibration and potential scaling bias, fixed-intercept (zero) linear regression was performed with the measurement from the proposed device regressed on the reference value; the resulting slope (β) reflects multiplicative (scaling) agreement, with $\beta \approx 1$ indicating minimal scaling bias. Lin's concordance correlation coefficient (CCC) was calculated to summarize overall agreement, incorporating both precision

and accuracy relative to the identity line.^{24,25} Proportional bias was evaluated by regressing the inter-device difference (proposed – reference) on their mean; a statistically significant non-zero slope indicates magnitude-dependent bias. Systematic bias in the mean was additionally tested using paired two-sided t-tests.

For a clinically oriented description of agreement, total deviation indices (TDI)^{26,27} were computed as the 90th (TDI90) and 95th (TDI95) percentiles of the absolute inter-device difference, representing the maximum difference expected to be exceeded in only 10% and 5% of eyes, respectively. Equivalence between devices was examined using the two one-sided tests (TOST) procedure²⁸ with pre-specified margins (± 0.1 , ± 0.2 , and ± 0.3 mm for ACD; ± 40 , ± 60 , and ± 80 μm for CCT), chosen to reflect approximately 1–2 pixels at the corneal plane and clinically acceptable differences for screening-level pachymetry. Equivalence was concluded when both one-sided tests rejected the null hypothesis that the mean inter-device difference exceeded the specified margin at $\alpha = 0.05$. Unless otherwise stated, a two-sided p-value < 0.05 was considered statistically significant. All analyses were performed in Python using NumPy, SciPy, and custom analysis scripts.

Real-time inference speed on Jetson Nano Orin: To evaluate the computational feasibility of on-device AI analysis, we profiled the lightweight LWBNA-unet on an NVIDIA Jetson Orin Nano developer kit configured to 15 W power mode. Slit-scan image acquisition was first completed over approximately 15 s per eye, after which all recorded

frames were processed sequentially by the AI pipeline. Processing a typical 51-frame scanning-slit video (480×720 resolution, 3 channels) required an additional 18.5 s end-to-end (≈ 2.7 fps), with per-frame inference latency ranging from 62 to 150 ms (a representative processing log is shown in Supplementary Fig. S1). Thus, data acquisition and AI processing occur in two distinct stages rather than concurrently, yielding a total acquisition-plus-processing time of approximately 33–35 s per eye. These results confirm that the model runs entirely on a low-power (<15 W) edge-AI module without cloud connectivity, supporting truly portable, battery-operated deployment.

As mentioned above, the program automatically exports all intermediate and final results for transparency and downstream analysis. For each eye, three key raw frames are saved—"cornea_frame" (corneal-centered), "iris_frame" (iris-centered) and "lens_frame" (lens-centered)—along with a composite "Results image" that overlays the segmentation masks and prints the computed biometric values (ACD in mm and CCT in μm). In parallel, a CSV file logs one row per case/ID with frame indices, CCT and ACD values from each selected frame, corneal diameter (pixels) used for scaling, and total processing time. This design makes the measurements reproducible (primary frames explicitly identified), auditable (segmentation overlays archived), and immediately usable for statistical analysis or integration into teleophthalmology dashboards for remote consultation.

Results

To evaluate the diagnostic performance of the portable scanning slit-light device integrated with AI-based image analysis, screenings were conducted on both healthy participants and patients with diverse anterior segment pathologies. The device consistently produced high-contrast slit images (Fig.4) that revealed clinically relevant ocular structures and abnormalities, enabling automated segmentation and parameter extraction. Figure 4(a) illustrates a cataract, where the lens exhibits a yellowish, partially opaque appearance consistent with advanced opacification. Figure 4(b) shows a narrow anterior chamber angle, a hallmark of primary angle-closure glaucoma. Figure 4(c) depicts corneal opacity, with pronounced light scatter indicating stromal irregularity, while Figure 4(d) highlights keratoconus, where corneal thinning and conical protrusion distort the slit reflection pattern.

Importantly, participants reported that imaging with the portable device was comfortable, with minimal glare or photic discomfort compared with conventional slit-lamp examinations. This was attributed to the device's lower-intensity, narrow slit illumination and brief scanning duration. These findings highlight that the system can capture subtle corneal and lenticular abnormalities essential for early disease detection, while maintaining a patient-friendly imaging experience with minimal

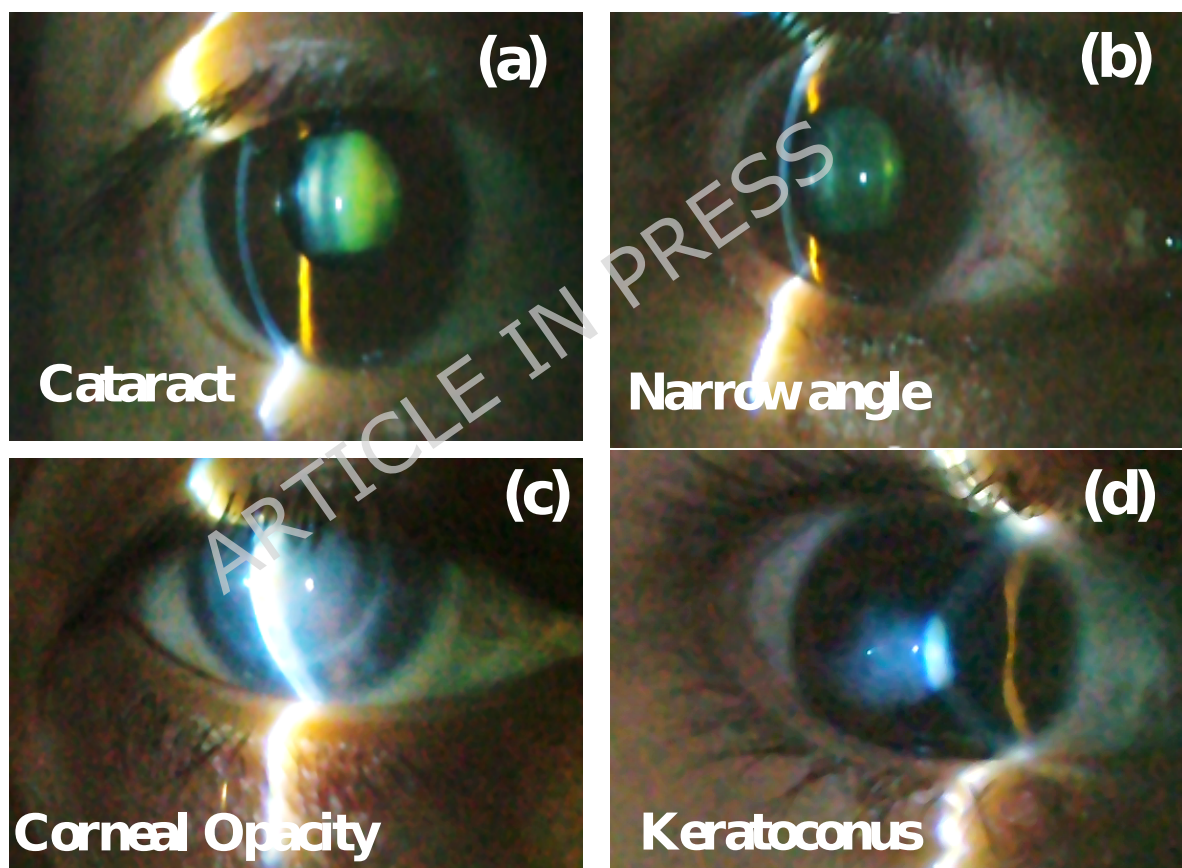


Figure 4. Representative anterior-segment images captured with the portable scanning slit-light device. (a) cataract, showing yellowish lens opacity; (b) narrow anterior chamber angle, characteristic of primary angle-closure glaucoma; (c) corneal opacity with increased light scatter indicating stromal irregularity; and (d) keratoconus -distortion of the slit reflection.

glare or discomfort—supporting its suitability for community screening, teleophthalmology, and repeated examinations.

Segmentation of anatomical features in scanning slit-lamp frames

Figure 5 shows the training and validation performance of the LWBNA-unet model in terms of the Dice similarity coefficient (DSC) across epochs. The training curve (black) showed a rapid increase during the initial epochs, surpassing 0.85 within the first 10 epochs and stabilizing above

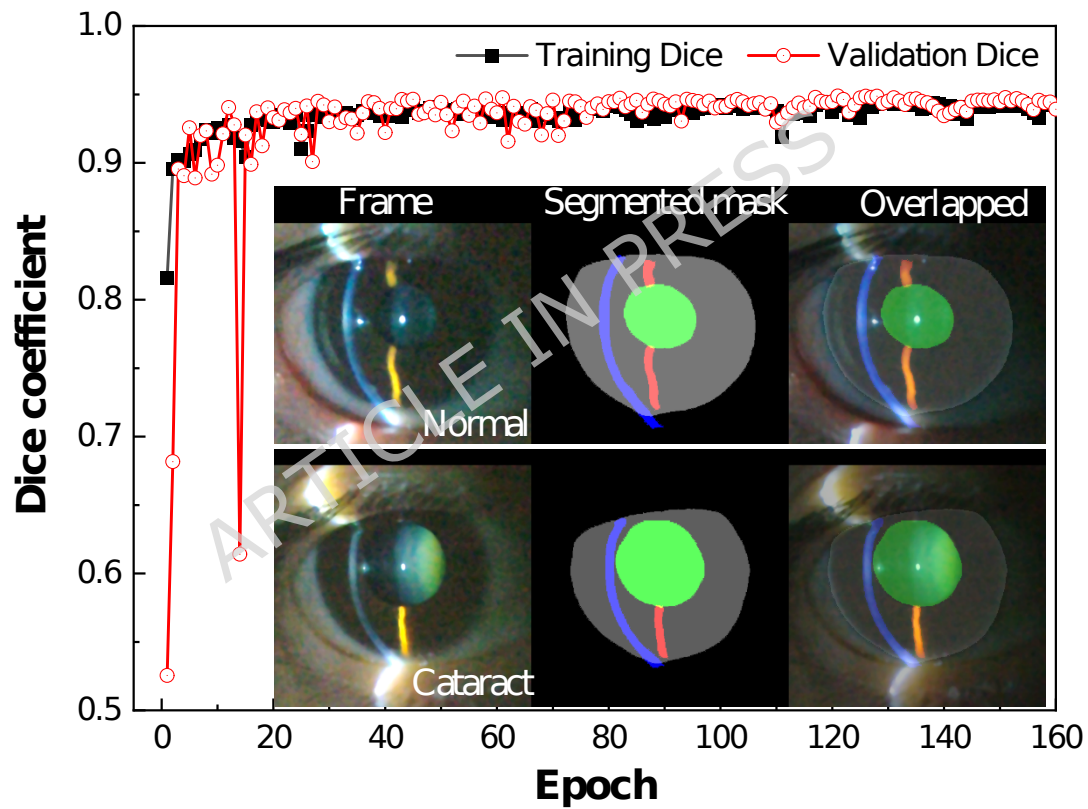


Figure 5. Training performance and representative segmentation results of the LWBNA-unet model. Training (black) and validation (red) Dice similarity coefficient (DSC) curves for the LWBNA-unet for segmentation of frames captured from a portable scanning slit-light device. Insets illustrate representative examples: (left) input frames from a normal eye and a cataract case, (middle) predicted segmentation masks highlighting corneal reflections (blue), iris reflections (red), pupil boundaries (green), and outer corneal surfaces (gray), and (right) overlays of model predictions on the original frames.

0.90 thereafter. The validation curve (red) followed a similar trajectory, though with fluctuations between 0.60 and 0.90 during the first 20 epochs, reflecting variability in early optimization. After ~30 epochs, the validation DSC stabilized consistently in the range of 0.92–0.95, indicating strong agreement between predicted segmentation masks and ground-truth annotations. Transient drops observed in the early phase were comparable to those previously reported for LWBNA-unet when segmenting fine anatomical structures such as the FAZ boundary.²¹ At peak performance, the model achieved a training DSC of 0.936 and a validation DSC of 0.947, demonstrating excellent agreement with expert annotations. These findings confirm that the model maintained a favorable balance between training and validation, with no evidence of overfitting, thereby validating its robustness for automated segmentation of frames captured with a portable scanning slit-light device.

The inset in Figure 5 further illustrates model performance. The first column shows representative input frames of a normal eye and a cataract case. The second column presents the corresponding segmentation masks generated by the model, where corneal reflections (blue), iris reflections (red), pupil boundary (green), and outer corneal surface (gray) are automatically delineated. The third column demonstrates the overlap between predicted masks and the original frames, highlighting the precise alignment of segmented structures with ocular anatomy. Together, these visual results reinforce the quantitative performance metrics.

Representative clinical examples

The algorithm consistently extracted one optimal frame for each quantitative parameter, providing standardized inputs for downstream analysis. Figure 6 shows representative outputs from this automated extraction and segmentation pipeline, alongside reference anterior-segment optical coherence tomography (AS-OCT; CASIA-2, Tomey) images for comparison. Fig. 6(a-e) correspond to a cataract case, and Fig. 6(f-j) depict a narrow-angle eye. Fig. 6(a-c) and Fig. 6(f-h) show the original frames automatically selected from the scanning-slit videos, while Fig. 6(d-e) and Fig. 6(i-j) illustrate the corresponding segmentation overlays generated by the LWBNA-unet model. The predicted contours accurately

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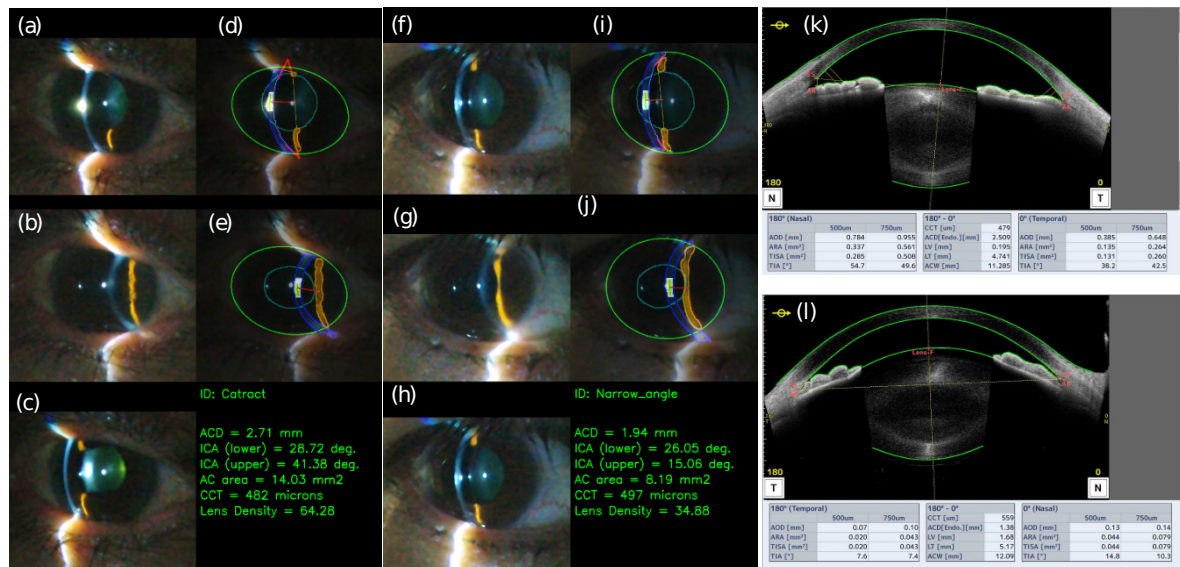


Figure 6. Automated frame selection, segmentation, and quantitative analysis using the portable scanning slit-light device compared with AS-OCT.

Representative results for two subjects captured with the portable device and analyzed by the LWBNA-unet model. **(a-e)** Cataract case and **(f-j)** narrow-angle case. Figs. (a-c, f-h) show the original frames automatically selected from scanning-slit videos; panels (d, e, i, j) display the corresponding segmentation overlays, where cornea (blue), iris (orange), pupil (green), and fitted corneal ellipse (light green) are delineated for geometric computation. Device-derived parameters are annotated in (c, h). **(k, l)** Reference AS-OCT; CASIA2, Tomey delineate the cornea, iris, pupil, and lens boundaries, enabling geometric derivation of anterior-segment parameters such as ACD, iridocorneal angle (ICA), anterior-chamber (AC) area, CCT, and lens-opacity index.

Quantitatively, the device-derived parameters clearly distinguished the two clinical phenotypes. In the cataract case, the anterior chamber was deeper (ACD = 2.71 mm), the iridocorneal angle wider (ICA = 41.4°), and the AC area larger (14.0 mm²), whereas in the narrow-angle eye, these values were markedly reduced (ACD = 1.94 mm; ICA = 15.1°; AC area = 8.2 mm²). These trends closely matched AS-OCT findings [Fig. 6 (k) & (l)],

where the corresponding CASIA-2-derived ACD and ICA values were 2.74 mm / 38–54° for the cataract eye and 1.66 mm / 7–15° for the narrow-angle eye. Additionally, the lens-density index [Fig. 6(c)&(h)] further differentiated the two conditions. The cataract case exhibited substantially higher lens density (64.3), consistent with lenticular opacity, whereas the younger, narrow-angle subject without cataract showed a much lower value (34.9). This highlights the system's ability to extract both geometric and photometric biomarkers from a single scanning-slit acquisition, enabling simultaneous assessment of angle-closure risk and lens transparency.

Minor discrepancies in absolute parameter values between modalities likely arise from differences in scanning geometry and measurement definition. The portable scanning slit-light device measures along an oblique illumination path, whereas AS-OCT acquires cross-sectional data in a fixed meridian, and ICA values vary depending on the reference location (typically 500 μm or 750 μm from the scleral spur). In addition, pixel-to-metric scaling in our device used a fixed corneal-diameter reference (12 mm), while AS-OCT employs built-in axial calibration. Despite these methodological differences, the close proportional agreement and consistent physiological trends confirm that the portable system achieves clinically reliable, multimodal quantification of anterior-segment structures.

Figure 7 further demonstrates the system's capability in a patient with keratoconus. Figure 7(a) shows a representative image frame

captured with the portable scanning slit-light device. The corresponding segmentation mask generated by the AI model, overlaid on the same frame, is shown in Figure 7(b). The segmented curvature clearly reproduces the paracentral corneal steepening characteristic of keratoconus, closely matching the corneal contour observed in the AS-OCT image (Fig. 7d). The same corneal opacity visible in the commercial slit-lamp photograph (Fig. 7c) is also evident in the portable device image, confirming high optical fidelity despite the compact hardware. Additional examples in the Supplementary Figures further demonstrate this consistency: the device accurately reproduces slit-beam features seen in commercial systems,

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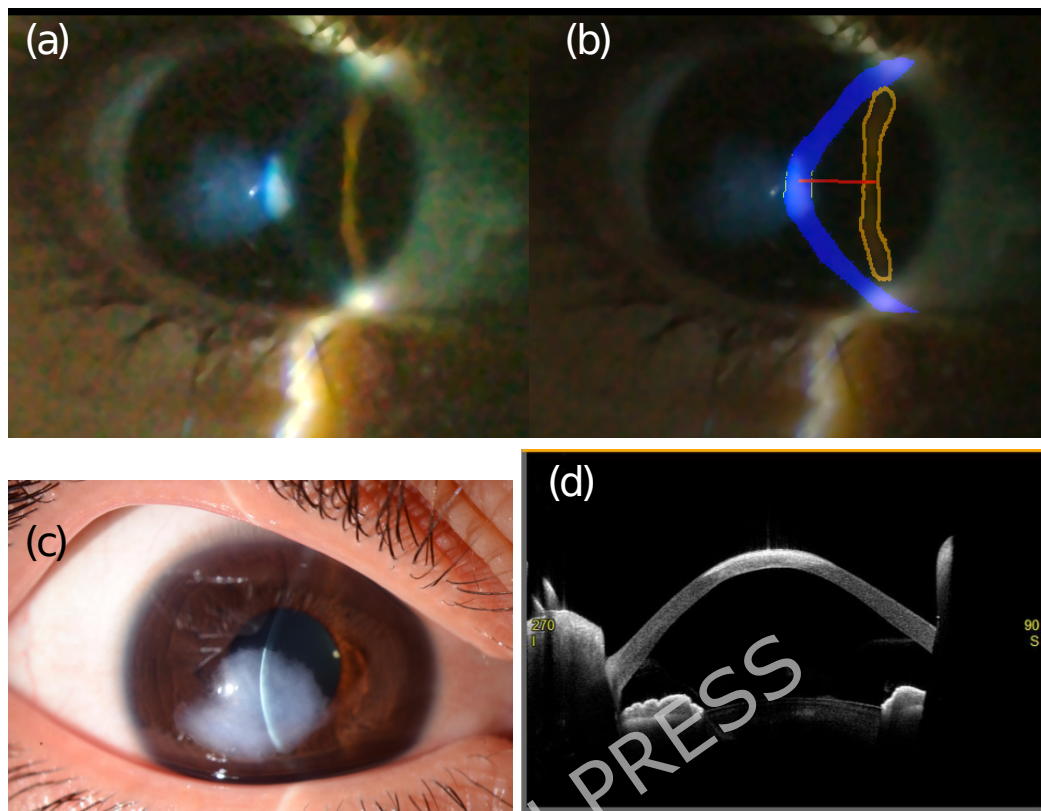


Figure 7. Detection of corneal curvature abnormality and opacity in keratoconus using the portable scanning slit-light device.

Representative case demonstrating the ability of the AI-assisted portable slit-light system to visualize and quantify corneal irregularity. **(a)** Frame automatically extracted from a scanning-slit video. **(b)** Corresponding segmentation overlay produced by the LWBNA-unet model, delineating the anterior corneal surface (blue). The curved corneal contour clearly depicts paracentral steepening characteristic of keratoconus. **(c)** Commercial slit-lamp photograph of the same eye showing localized corneal opacity, which is also evident in the portable-device image. **(d)** Reference anterior-segment OCT (CASIA2, Tomey) cross-section including pseudophakia without a crystalline lens and posterior capsule reflections (Supplementary Figures S2-S3). Collectively, these results show that the portable scanning slit-light device, combined with the lightweight deep-learning model, can capture and quantify corneal and lenticular abnormalities with fidelity comparable to clinical slit-lamp

imaging—while enabling quantitative analysis not possible with conventional devices.

Validation of ACD and CCT measurements by AS-OCT and the portable scanning slit-light device

Building on the previous section, where the scanning slit-light device was shown to qualitatively and quantitatively capture anterior chamber configurations, we next validated its quantitative measurements against the clinical reference standard, AS-OCT. Because the CASIA-2 provides direct measurements of corneal diameter in addition to ACD and CCT, the pixel-based outputs of the portable scanning slit-light device were scaled using each participant's actual corneal diameter rather than a fixed nominal value of 12 mm. This calibration step ensured accurate conversion from pixels to micrometers and was particularly important for CCT, where small deviations in scaling can substantially affect measurement precision due to its thinner profile compared with ACD. In a complementary analysis, we also applied a fixed 12-mm dynamic corneal scaling to all eyes, allowing direct comparison between an individualized (CASIA-adjusted) calibration and a simple fixed-diameter approach. Agreement analyses were performed as described in the Statistical analysis section. Figure 8 summarizes the comparative analysis of ACD and CCT for the case of CASIA-adjusted corneal diameter.

For ACD, agreement between the portable device and CASIA-2 was consistently high under both scaling strategies. With CASIA-adjusted corneal diameter scaling, the bias was 0.044 ± 0.175 mm (95% LoA -0.300

to +0.388 mm; CR 0.344 mm; paired $p = 0.08$). Linear association and concordance were strong (Pearson's $r = 0.916$, CCC = 0.903), and the fixed-intercept slope was close to unity ($\beta = 0.985$; 95% CI 0.967–1.003). A small proportional bias was detected (slope -0.137 , $p = 0.028$). Overall, 44%, 72%, and 88% of eyes were within ± 0.1 , ± 0.2 , and ± 0.3 mm, respectively, and TDI90/TDI95 were 0.313/0.375 mm. Interestingly, with a fixed 12-mm dynamic scaling using corneal diameter gave very similar or slightly better results (bias 0.008 ± 0.164 mm; 95% LoA -0.313 to $+0.329$

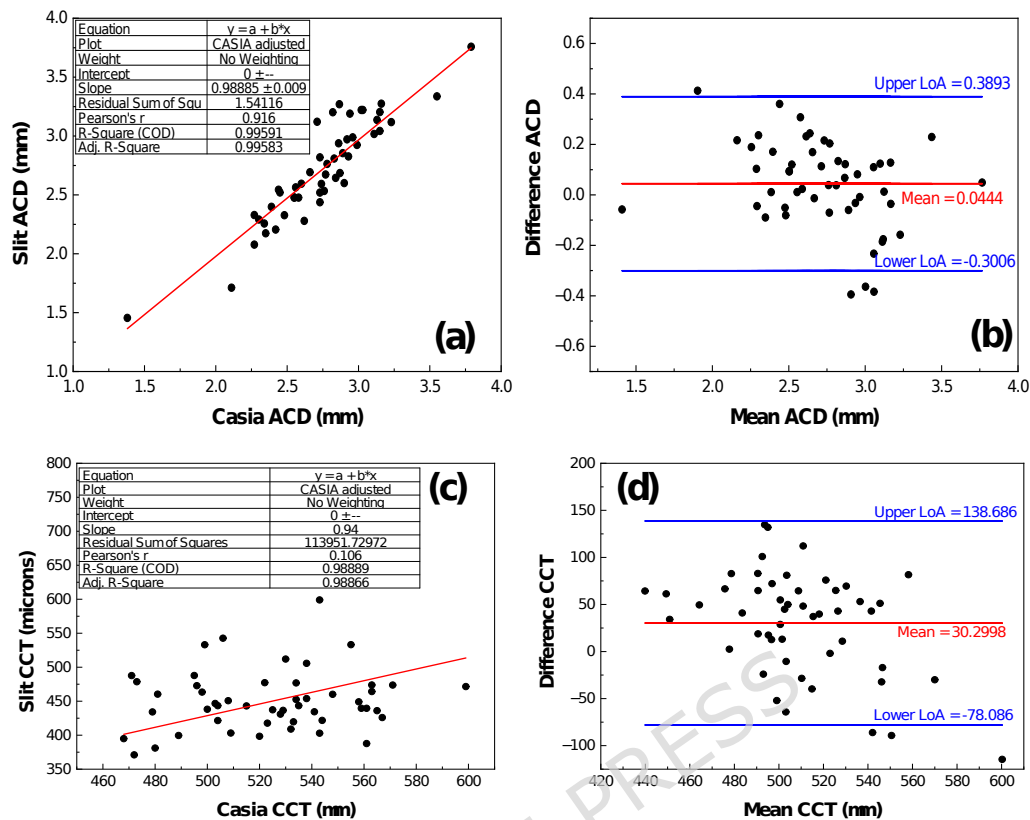


Figure 8. Validation of anterior chamber depth (ACD) and central corneal thickness (CCT) measurements between AS-OCT and the portable scanning slit-light device.

Comparison of quantitative parameters obtained using the AI-enhanced portable scanning slit-light system and anterior-segment optical coherence tomography (AS-OCT; CASIA-2, Tomey). (a) Correlation between ACD values measured by the two modalities (Pearson's $r = 0.916$), showing near-identity proportional agreement. (b) Bland-Altman plot of ACD measurements, indicating minimal bias (+0.044 mm) and narrow 95% limits of agreement (−0.301 to +0.389 mm). (c) Correlation between CCT values ($r = 0.106$), demonstrating strong linear association across devices. (d) Bland-Altman plot for CCT, showing a small positive bias (+30.3 μm) with limits of agreement mm; CR 0.321 mm; paired $p = 0.73$; $r = 0.913$; CCC = 0.912; $\beta = 0.996$, 95% CI 0.980–1.013) with no significant proportional bias (slope −0.048, $p = 0.44$). In this configuration, 52%, 74%, and 90% of eyes were within ± 0.1 , ± 0.2 , and ± 0.3 mm, respectively, and TDI90/TDI95 were

0.282/0.340 mm. TOST demonstrated equivalence for all three ACD margins (± 0.1 , ± 0.2 , and ± 0.3 mm) for both scaling methods.

For CCT, agreement was more modest. With CASIA-adjusted scaling, the portable device underestimated CCT by 30.3 ± 55.3 μm on average (95% LoA -78.1 to $+138.7$ μm ; CR 108.4 μm ; paired $p = 0.00032$). Correlation and concordance with CASIA-2 were weak (Pearson's $r = 0.11$, CCC = 0.076), although the fixed-intercept slope remained close to 1 ($\beta = 0.94$; 95% CI 0.91–0.97). Proportional bias was significant (slope -0.78 , $p = 0.0018$). Only 36%, 58%, and 78% of eyes were within ± 40 , ± 60 , and ± 80 μm , respectively, and TDI90/TDI95 were 90.4/113.5 μm . TOST did not support equivalence at ± 40 μm but did at ± 60 and ± 80 μm . With 12-mm dynamic scaling, the mean bias was slightly smaller (22.4 ± 55.7 μm ; 95% LoA -86.8 to $+131.6$ μm ; CR 109.2 μm ; paired $p = 0.0065$), but correlation and concordance remained low ($r = 0.17$, CCC = 0.134; $\beta = 0.95$, 95% CI 0.93–0.98) and proportional bias persisted (slope -0.82 , $p = 0.00045$). TDI90/TDI95 were 90.2/100.5 μm , and 44%, 64%, and 78% of eyes were within ± 40 , ± 60 , and ± 80 μm , respectively. In this case, TOST indicated equivalence for all three CCT margins.

Taken together, these results show that ACD measurements from the portable scanning slit-light device agree closely and are statistically equivalent to CASIA-2, with absolute differences typically ≤ 0.3 mm, especially when using 12-mm dynamic corneal scaling—supporting near-interchangeable use for screening. In contrast, CCT estimates are biased by ~ 20 – 30 μm on average, with wide LoA and significant proportional bias.

Given that one image pixel corresponds to $\approx 40\text{ }\mu\text{m}$ at the corneal plane (so a 1-2-pixel discrepancy already yields 40–80 μm of apparent CCT difference), much of this scatter likely reflects the spatial-resolution limit of the current prototype rather than a fundamental flaw in the measurement principle. At present, CCT from the portable device should therefore be considered exploratory and not a direct substitute for CASIA-2 pachymetry in individual clinical decision-making.

Discussion

This study presents an AI-enhanced portable scanning slit-light device capable of delivering quantitative anterior-segment biometry at a material cost below USD 500 while operating entirely on-device. In particular, the prototype provides ACD measurements that show close agreement and statistical equivalence with AS-OCT, together with exploratory estimates of CCT derived from the same scan. Such capability directly addresses the global need for affordable, objective screening tools for primary angle-closure glaucoma and other anterior-segment disorders—conditions that remain underdiagnosed, especially in Asia, where shallow anterior chambers and corneal thickness are rarely measured in community settings.

Conventional slit-lamp biomicroscopy remains indispensable in ophthalmic practice but is inherently qualitative and operator-dependent. Classical techniques such as the Smith and Redmond-Smith methods, or the Haag-Streit optical pachymeter, demonstrated that ACD and CCT could in principle be estimated at the slit lamp.²⁹ However, these are

manual, single-point measurements requiring the examiner to align the beam, adjust the graticule, and record values individually. They depend on a fixed head position and assume uniform illumination without real-time correction for oblique incidence. In practice, the largest source of variability in Smith-style readings arises from operator-dependent alignment; even a 2 mm change in patient position can alter results. In the original Smith method, for instance, slit length is adjusted until the corneal and iris or lens images meet, and ACD is inferred from this length using fixed geometry—an approach that neglects optical distortion and dynamic alignment errors. Unlike these classical Smith-type slit-lamp methods, which rely on a universal proportionality constant between the apparent slit image and depth, the proposed approach estimates ACD directly from the segmented corneal and iris interfaces and applies per-eye scaling based on the segmented corneal diameter, without relying on any universal geometric factor. Because the corneal diameter is re-estimated in each accepted frame, modest head-position-related z-motion primarily appears as a change in magnification that is automatically compensated by this dynamic, per-frame scaling. Although inter-subject variability in corneal diameter exists (typically ~11.5–12.3 mm),^{30,31} this introduces only a bounded proportional scaling uncertainty. Under worst-case assumptions, the resulting error in absolute measurements remains below approximately $\pm 3\text{--}4\%$, which is acceptable for screening-level biometry and does not affect relative comparisons within the same eye.

Hand-held and smartphone-attachable slit-lamps improve accessibility but still rely on manual scanning, variable illumination and

focus, and lack validated quantitative outputs. Caruso et al.¹³ described a 3D-printed smartphone slit-lamp producing high-resolution images, yet analysis remained qualitative and observer-dependent. Xu et al.¹⁴ developed a deep-learning framework for image-quality classification in slit-lamp and smartphone photos, underscoring that image quality itself is a major bottleneck for automated analysis. Qian et al.³² trained a modified ResNet-34 to predict ACD from single iPhone photographs captured under diffuse illumination, achieving only moderate correlation ($r \approx 0.63$) with Lenstar and larger errors in shallow ACD cases, highlighting the limitations of inferring depth from uncalibrated 2D images. Shimizu et al.³³ subsequently demonstrated deep-learning prediction of ACD from smartphone-based slit-lamp videos, yet their pipeline remained dependent on manual phone alignment and lacked per-frame scale calibration. Similarly, a recent *npj Digital Medicine* report¹⁰ used eight morphological parameters from single smartphone-slit images for narrow-angle detection but showed reduced performance on external validation, likely due to variable optical alignment and absent path correction. The Smart Eye Camera confirmed that smartphone converters can support teleophthalmology, though they still require manual grading and ruler-based calibration.

To address these limitations, we designed the portable scanning slit-light device specifically to remove as much operator dependence and geometric uncertainty as possible. Instead of relying on single, manually acquired photographs with uncontrolled alignment and unknown scale, the device captures standardized scanning-slit videos under fixed

geometry and performs fully automated frame selection, segmentation, and calibration. A motorized scanning mirror with regulated illumination maintains a fixed optical path, while the acquisition software automatically selects cornea- and iris-centered frames, minimizing operator bias and ensuring consistent alignment. The lightweight LWBNA-unet performs real-time segmentation of corneal reflections, iris reflections, pupil boundaries, and corneal surfaces. To obtain true anatomical dimensions, the system first recenters each frame so that the pupil lies at the image center, then applies slit-incidence-angle correction and per-eye scaling using each participant’s actual corneal diameter. Unlike classical Smith or Redmond-Smith methods²⁹ based on fixed-angle approximations and manual readings, our model accounts for spatially varying local angles $\theta(x)$ across the slit, compensating for oblique illumination in every frame. Segmented corneal boundary is fitted with an ellipse to estimate the effective corneal diameter, providing a frame-specific scaling factor that compensates for variation in eye size, image magnification, and camera distance. This geometric correction and adaptive scaling transform segmentation masks into calibrated ACD (mm) and CCT (μm) measurements. In our validation, ACD showed strong agreement and statistical equivalence with AS-OCT (small bias, narrow LoA, high CCC), whereas CCT exhibited larger scatter, wider LoA, and low concordance, consistent with the resolution limits of the current prototype rather than full tomographic correction.

Small offsets between modalities are expected—even among “gold-standard” clinical instruments—because measurements depend on

illumination geometry, segmentation algorithms, calibration, and the exact metric definition. For example, CASIA-2 and the IOLMaster 700 show statistically significant between-device differences for several anterior-segment parameters with non-trivial 95% LoA, including CCT (-30.06 to $+0.43$ μm) and ACD (-0.03 to $+0.24$ mm), despite strong correlations overall.³⁴ Likewise, two swept-source AS-OCTs (CASIA-2 and ANTERION)³⁵ agree strongly in trend yet still exhibit measurable between-instrument differences. Doors et al.³⁶ found that Visante OCT, Orbscan, and Pentacam showed systematic differences in both CCT and ACD despite good overall reliability, and recommended caution when pooling measurements across devices. Similar findings have been reported for Pentacam versus Lenstar and Visante OCT, and for Pentacam versus AS-OCT in dry eye and refractive surgery cohorts.^{37,38} In this context, the ACD bias observed with our device (~ 0.0 – 0.04 mm) is comparable to inter-device differences reported between established systems, whereas the CCT bias (~ 20 – 30 μm) is within the range of known pachymetry offsets but accompanied by wider LoA due to coarser spatial sampling. Residual discrepancies are also consistent with differing sampling geometries: our oblique slit section (with incidence-angle correction) versus OCT's cross-sectional meridians; definition sensitivity for ICA (e.g., 500 vs 750 μm from the scleral spur); and our pixel-to-metric scaling from corneal diameter (robust for ACD but more sensitive for thin structures like CCT) versus OCT's axial calibration. We therefore interpret ACD from the portable device as screening-grade and near-interchangeable with AS-OCT, while CCT should currently be viewed as a secondary, resolution-limited metric.

An additional and clinically relevant advantage of the proposed system is its ability to acquire full-color visible-light images of the anterior segment during scanning. Unlike AS-OCT systems such as CASIA, which operate in the near-infrared spectrum and produce grayscale cross-sectional images, or Scheimpflug-based devices such as the Pentacam, which rely on monochromatic blue illumination for tomographic reconstruction, the portable scanning slit-light device captures true color reflectance from the cornea, iris, and crystalline lens. This enables direct visualization of clinically important features—such as cortical and nuclear lens opacities, anterior capsular changes, and pigment variations—that are difficult or impossible to assess using grayscale OCT or blue-light Scheimpflug imaging alone. Because cataract diagnosis and grading depend strongly on wavelength-dependent scattering and absorption most apparent under visible-light illumination, preserving natural color is particularly advantageous. In this way, a single scanning-slit video can provide both quantitative biometry (e.g., ACD, CCT, ICA) and qualitative or AI-driven assessment of lens opacity and anterior-segment appearance, reducing reliance on multiple dedicated instruments.

This multimodal capability—combining geometric measurements with color-based tissue characterization—positions the system primarily as a screening tool rather than a replacement for OCT or Scheimpflug tomography. Whereas commercial slit-scanning and swept-source OCT devices such as Pentacam HR and ANTERION are optimized for high-precision biometry and refractive surgery planning at a cost of tens of thousands of dollars, the present prototype targets a different point in the

design space: an ultra-low-cost, handheld device that provides screening-level ACD measurement together with color-based evaluation of cataract and angle anatomy from a single scan. The observed agreement with AS-OCT should therefore be interpreted as evidence of screening-level reliability for ACD, rather than full equivalence to high-resolution tomographic systems, particularly for CCT. In resource-limited or high-throughput screening environments, this combination of low cost, handheld form factor, and multimodal output provides a practical and clinically meaningful advantage.

Because the device captures a full scanning-slit video rather than a single static frame, it can extend far beyond ACD and CCT measurement to functionalities that traditionally require multiple instruments. From the same acquisition session, the pipeline can support Van Herick ratio grading for angle-closure risk, objective lens-opacity indices for cataract evaluation, dynamic pupillometry for autonomic-function assessment, and corneal-curvature or angle-geometry mapping for ectasia screening. Supplementary Figure S4 illustrates this qualitative angle assessment using representative normal [Fig. S4(a), (b)] and angle-closure eyes [Fig. S4(c), (d)]. In the normal eye, the temporal-limbal view with a fixed-width slit beam clearly shows a distinct band of anterior chamber between the corneal endothelium and the anterior iris, with well-separated corneal and iris reflections. In the angle-closure eye, the same temporal-limbal view reveals a much thinner bright band and a crowded peripheral iris, consistent with a shallow peripheral anterior chamber and increased risk of angle closure. This qualitative impression matches with a wider ACD for

the normal eye and a relatively shallow for the angle-closure case in the iris-centered frames. These results demonstrate that the current device provides not only quantitative biometry but also diagnostic-quality visualization of anterior-segment anatomy, enabling multi-parameter screening from a single, rapid scan. Moreover, because each frame is automatically aligned, centered, and geometrically scaled, the resulting dataset forms a standardized, high-quality foundation for anterior-segment AI—supporting models that can progress from simple “normal vs abnormal” detection toward multimodal risk stratification across diverse populations and devices.²²

By contrast, most prior AI approaches have relied on single static slit-lamp or smartphone photographs acquired under uncontrolled illumination and variable geometry. Such images are suitable only for coarse classification and lack the spatial consistency, calibration, and repeatability required for quantitative analysis or reliable cross-site deployment. As a result, these models often show substantial performance degradation on external datasets—particularly for subtle features such as narrow angles, early lens opacities, or mild ectasia. In our system, geometry-controlled scanning videos and per-frame calibration provide a stable input for AI, while on-device inference on a Jetson Orin Nano (51-frame scan processed in 18.5 s; ≈ 2.7 fps) allows fully offline, battery-powered operation without cloud connectivity. This combination of calibrated video-based measurements, diagnostic feature extraction, and standalone deployment makes the prototype a practical candidate for

teleophthalmology and community screening in environments where AS-OCT, gonioscopy, or high-end slit-lamps are not available.

Future work will extend this platform beyond quantitative ACD and CCT measurement toward a fully AI-enabled anterior-segment screening system. Although the present study demonstrates the device's ability to capture diagnostic features, large-scale validation of screening performance will require curated, multi-site datasets collected from community programs, hospitals, clinics, and population-based surveys. Because the device already generates standardized, geometry-controlled video data, these future datasets can support deep-learning models capable of automatically detecting and grading major anterior-segment diseases—including primary angle-closure disease (using Van Herick ratios, ACD, and angle metrics), cataract (via lens-opacity indices), keratoconus and other ectatic disorders (from peripheral curvature and thinning patterns), and autonomic dysfunction (via dynamic pupillometry). The same acquisition and segmentation framework may be expanded to compute additional biometric parameters (e.g., AOD500, TISA500, etc.) and to combine multiple biomarkers into unified risk-stratification scores suitable for teleophthalmology dashboards. Together, these developments will evolve the device from an ultra-low-cost imaging tool into a scalable platform for next-generation, AI-driven anterior-segment diagnostics with the potential to transform community eye care globally.

Limitations

This study has several limitations. First, the sample size was modest and derived from a single hospital and one community eye-screening event, which may limit generalizability. Although video-based acquisition increased the number of analyzable frames, larger multi-center datasets—including broader ethnic diversity and a wider range of anterior-segment conditions—will be essential for robust AI development and external validation. Second, all participants were Japanese. Even with per-eye geometric scaling, inter-ethnic differences in ocular anatomy—such as corneal diameter, anterior-chamber configuration, and iris pigmentation—may influence performance. Broader demographic representation will therefore be required to ensure applicability across global populations. Third, although the system demonstrated reliable frame selection and segmentation in controlled environments, real-world deployment may encounter additional challenges such as motion blur, blinking, poor alignment, or media opacities that were not fully represented in the present cohort. Prospective, field-based usability studies will be necessary to evaluate robustness under routine screening conditions.

Fourth, central corneal thickness estimation is currently constrained by the spatial resolution of the prototype. In the present design, the effective lateral sampling at the corneal plane is on the order of $\sim 40\text{ }\mu\text{m}$ per pixel, so even a 1-2-pixel segmentation error can translate into 40–80 μm of apparent CCT difference, consistent with the observed TDI90/TDI95 and wide limits of agreement. Future improvements using higher-resolution image acquisition (reduced down-sampling) and more powerful

edge hardware (e.g., higher-tier Jetson modules capable of processing full-resolution videos in real time) could substantially improve CCT precision and potentially shorten acquisition and processing times, without compromising portability.

Despite these limitations, the device's extremely low cost, operator-independent acquisition, and ability to perform AI-assisted quantitative analysis—as well as detect disease-related features—highlight its potential for community screening, teleophthalmology, and deployment in resource-limited settings.

Conclusions

This work introduces the first ultra-low-cost, AI-integrated portable scanning slit-light device capable of delivering quantitative anterior-segment biometry with accuracy comparable to AS-OCT. By combining geometry-controlled optical scanning, automated frame selection, deep-learning-based segmentation, and per-eye anatomical calibration, the system generates reproducible measurements of ACD and CCT and simultaneously captures disease-related features such as cataract opacity, narrow angles, and corneal ectasia from a single video acquisition. The device performed reliably across diverse clinical presentations while maintaining excellent patient comfort, demonstrating its suitability for repeated use and community deployment. With material costs below USD 500 and fully on-device inference, the platform overcomes the mobility, cost, and expertise barriers inherent to conventional slit-lamps and AS-

OCT systems. These results establish a new class of accessible, quantitative, operator-independent anterior-segment imaging devices. Future expansion toward AI-based disease classification, multi-parameter risk stratification, and large-scale screening will position this technology as a scalable foundation for teleophthalmology and global eye-health initiatives, especially in resource-limited settings.

Data Availability

The clinical imaging data used in this study are not publicly available due to participant privacy restrictions and institutional ethical guidelines. De-identified data may be made available from the corresponding author upon reasonable request and with approval from the Tohoku University Clinical Research Ethics Committee.

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Contributions

N.K., P.S., and T.N. conceived and supervised the study. N.K. developed the optical system and device hardware. P.S. assembled the Jetson AI module, implemented control software, and developed the AI models in collaboration with N.K. T.M., N.K., N.H., M.S., and S.T. acquired patient data using the portable scanning slit-light device and AS-OCT. All authors contributed to data interpretation, device refinement, and clinical validation at Tohoku University Hospital. N.K. and P.S. performed data analysis and co-wrote the manuscript with input from all authors and supervision from T.N.

Competing interests

The authors declare no competing interests.

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